

Vikas Thamizharasan

 [vikastmz.github.io](https://github.com/vikastmz)

 [@vikastmz](https://twitter.com/vikastmz)

 vthamizharas@umass.edu

EDUCATION

University of Massachusetts, Amherst

Ph.D. in Computer Science (2022 - *Current*)

Advisor: Prof. Evangelos Kalogerakis

Brown University

Masters in Computer Science (2021)

Advisors: Prof. James Tompkin and Prof. Daniel Ritchie

IIIT - Hyderabad

Bachelor Of Technology in Computer Science and Engineering (2018)

WORK EXPERIENCE

JUNE 2025 -	AI Research Scientist Intern , Autodesk Research	Remote
AUGUST 2025	Designed and trained generative models for CAD. Research published in ECCV'26.	
MAY 2024 -	Research Scientist Intern , Adobe Research	San Jose, CA
JAN 2025	Developed generative diffusion models for vector graphics, including VecFusion (CVPR'24).	
MAY 2023 -	Research Scientist Intern , Adobe Research	San Jose, CA
JAN 2024	Designed neural implicit representations for vector graphics, NIVeL (CVPR'24), and investigated training-free distillation methods on diffusion models.	
JAN 2022 -	Research Intern , Activision Blizzard	Los Angeles, CA
AUG 2022	Contributed to digital human research and development. Received credits for <i>Call of Duty: Modern Warfare II</i> (2022).	
MAY 2020 -	Graduate Research Assistant , Visual Computing Lab, Brown University	Providence, RI
DEC 2021	Researched problems in the intersection of CV, Graphics, and ML.	
MAY 2021 -	Programming Intern , Activision Blizzard	Remote
AUG 2021	Conducted R&D on statistical 3D face models for representing geometry and appearance.	
AUG 2018 -	Research Intern , INRIA	France
FEB 2019	Investigated deep learning methods for depth estimation and face attribute analysis from structured light data with the STARS team.	
MAY 2017 -	Google Summer of Code Intern , Google	Remote
AUG 2017	Contributed to ViSP CAD model conversion tools. [Source code and Wiki]	

PUBLICATIONS

BRepFacetGen: Reverse Engineering B-Reps By Generative Face Segmentation	ECCV 2026
K. H. Hui, V. Thamizharasan, P. K. Jayaraman, X. Xu	
[Paper] [Website]	
Mean-Shift Distillation for Diffusion Mode Seeking	WACV 2026
V. Thamizharasan, N. Chatzis, I. Georgiev, M. Fisher, D. Liu, N. Zhao, E. Kalogerakis, M. Lukác	
[Paper] [Website]	
NIVeL: Neural Implicit Vector Layers for Text-to-Vector Generation	CVPR 2024
V. Thamizharasan, D. Liu, M. Fisher, N. Zhao, E. Kalogerakis, M. Lukác	
[Paper] [Website]	
VecFusion: Vector Font Generation with Diffusion	CVPR 2024 Highlight
V. Thamizharasan*, D. Liu*, S. Agarwal, M. Fisher, M. Gharbi, O. Wang, A. Jacobson, E. Kalogerakis	
[Paper] [Website]	
GEM3D: GEnerative Medial Abstractions for 3D Shape Synthesis	SIGGRAPH 2024
D. Petrov, P. Goyal, V. Thamizharasan, V. Kim, M. Gadelha, M. Averkiou, S. Chaudhuri, E. Kalogerakis	
[Paper] [Website]	
Learning Physically-based Material and Lighting Decompositions for Face Editing	CVPR 2021 AICC, CVM 2022
Q. Zhang*, V. Thamizharasan*, J. Tompkin	
[Paper] [Presentation] [Code]	

PUBLICATIONS (continued)

Shape from Tracing: Towards Reconstructing 3D Object Geometry and SVBRDF Material from Images via Differentiable Path Tracing

3DV 2020

P. Goel, L. Cohen, B. Guesman, V. Thamizharasan, J. Tompkin, D. Ritchie

[\[Webpage \]](#) [\[Paper \]](#)

Face Attribute Analysis from Structured Light: An End-to-End Approach

Multimedia Tools and Applications 2019

V. Thamizharasan, A. Das, D. Battaglini, F. Bremond, A. Dantcheva

[\[Paper \]](#)

PROJECTS

Implicit Texture Fields for 3D Objects

Trained neural networks to represent 3D object textures as continuous fields, enabling texture reconstruction from an input mesh and a single image, as well as generative texture synthesis.

[\[Paper \]](#) [\[Code \]](#) PyTorch

Non-Linear Deep Face Models

Developed a generative model of facial geometry and appearance with disentangled identity and expression representations. Captured fine texture detail and supported semantic control of generated faces.

[\[Code \]](#) PyTorch

StyLit: CPU/GPU Stylization of 3D Renderings

Implemented CPU and GPU versions of StyLit, an illumination-guided method for transferring the visual style of an example image to 3D renderings.

[\[Code \]](#) [\[Video \]](#) C++, CUDA

Rendering, Geometry Processing, and Physical Simulation – CSCI 2240

Implemented a Monte Carlo path tracer, mesh subdivision, simplification and remeshing algorithms, and deformable-solid simulation using the finite element method.

[\[Path Tracer \]](#) [\[Geometry Processing \]](#) [\[Simulation \]](#) C++, Eigen

Single-Image 3D Reconstruction and Editing

Built an interactive editor that reconstructs 3D objects from a single image by fitting geometric primitives to user-specified constraints. Used the reconstructed geometry to support object manipulation within the image.

[\[Code \]](#) PyQt3D, OpenCV, SciPy

TECHNICAL SKILLS

LANGUAGES	Python, C++, C, MATLAB, C#, Bash, Javascript, CUDA, Racket/Scheme.
LIBRARIES	Pytorch, Tensorflow, OpenCV, SciPy, NumPy, Qt, Eigen, OpenGL, Windows Form App.
TOOLS	Blender, Inkscape, LaTeX, GCP, AWS, RunAI, Android Studios, Unity.

ACHIEVEMENTS

- 2017 Microsoft Code.Fun.Do Hackathon Winner Hyderabad, India.
- 2013 Top 5 in World Health Organization Art competition.

PROFESSIONAL SERVICES

Reviewer. NeurIPS (2025); ICML (2025); SIGGRAPH (2025); CVPR (2025, 2024); ECCV (2024).

Teaching Assistant, University of Massachusetts, Amherst.

Machine Learning, CSCI 589, Fall 2025.

Game Programming, CSCI 576, Fall 2022, 2023.

Intelligent Visual Computing, CSCI 674, Spring 2023, 2024.

Teaching Assistant, Brown University.

Topics in 3D Computer Vision and Machine Learning, CSCI2952K, Fall 2020.

Computer Vision, CSCI1430, Spring 2021.

Co-organizer, Machine Learning & Friends Lunch (MLFL), University of Massachusetts, Amherst.

Volunteer, IEEE International Conference on Image Processing, Applications and Systems.

Head of Art Committee, IIIT-Hyderabad.